Chapter 8

Informal teaching: More than an interpreter

In Chapter 7, I explained how learning and teaching lie at the core of several collaboration performances. Reflecting on the mass of data collected from our interviews and observations, and studying engineers working in four different countries, it is apparent that difficulties with learning and teaching lie behind most of the factors that frustrate collaboration performances by engineers.

Fortunately, recent intellectual advances in the learning sciences and education psychology have much to offer that can help reduce these frustrations for engineers by providing helpful guidance to improve learning and teaching. Most of the research has been directed at formal learning and education institutions, i.e. classroom learning situations. Collaboration in engineering, however, mostly relies on informal teaching. It should be noted, though, that the fundamental ideas exposed by research on formal teaching and learning are still applicable and can be very helpful.

In this chapter, I will outline some of the results that offer the greatest potential for engineering practice.

Another good reason to learn more about teaching is that only a minority of engineers are comfortable with reading in order to learn, perhaps less than 10%. Given that you have read this far in the book, you could be one of them. For readers and non-readers alike, this book holds a great deal of value: learning and practicing skills makes it easier for everyone involved in an engineering project. By passing on useful aspects of this book to engineers who are not as comfortable with reading, you will not only be able to practice your teaching, but also reinforce the ideas in your own mind and encourage others to read more for themselves.

Most of the research literature on workplace learning has focused on the need to improve the skills and knowledge of people working in an enterprise.[1] Other literature focuses on the development of specific capabilities, for example in communities of practice.[2] Recent research, however, has demonstrated that in many workplaces, people come together to collaborate for a limited period of time in relatively informal organisational structures, most of which are hardly formal enough to be regarded as 'teams'.[3] This can only happen with effective learning and teaching performances. Learning and teaching extend far beyond organisational skills and knowledge development activities and seem to be much more common in the workplace than has been previously acknowledged.

At the same time however, teaching with the aim of skills or knowledge improvement is an important aspect of engineering practice. As humans, we all age and progress

from being novices through learning throughout our working lives. Teaching others is a part of being human, whether it is our own children, our peers, novices, or even senior engineers who need to learn something new. Dianne Bailey and her colleagues, in an extensive and detailed study of novice engineers in several different workplaces, found that they were spending up to an hour or so every day, on average, in informal one-to-one teaching situations.[4] More senior engineers were doing the teaching, often spending this amount of time with two or three less experienced engineers, guiding their work and helping them learn more advanced skills. Often, as an engineer, you will be called upon to teach new knowledge to people with much more experience than yourself. Shortly, we will review some of the situations in which these issues present themselves.

First though, a few words about theories of learning and teaching. Developments in the last few decades have provided us with powerful ways to understand what happens when people learn. In learning about these theories however, many engineers may have to think differently about theories in the context of engineering. Theories of human behaviour are not like the familiar theories that have emerged from the traditional physical and engineering sciences. For a start, most are not expressible in mathematical form; instead, they rely on extensive written argumentation. The few theories with quantitative empirical support in sociology and psychology need to be interpreted for their results to be useful.

For most engineers, four significant thinking changes will be needed.

i) Social theories explain, but generally don't predict, human behaviour. Instead, their value lies in providing concepts and language that help you perceive what is happening. You can then discern patterns of behaviour and learn more quickly from experience.
ii) Instead of a single 'accepted' theory, for example, the theory of thermodynamics, the social sciences offer multiple theories and approaches, all of which may be valid in a given situation. All of them can be informative, although often they conflict with each other and the explanations that emerge from them can be contested. None offer a complete understanding, and the variety of different approaches offers a richer understanding of human behaviour than a single theory.
iii) By and large, most social theorists would now agree on the impossibility of truely independent, objective observations of human behaviour. As we briefly discussed in Chapter 7, everyone sees the same text or human situation through a different 'lens' that is shaped by their prior knowledge, experience, and beliefs. By understanding the ways that other people see a particular situation we can learn to 'place ourselves in their shoes',[5] and begin to appreciate the particular lenses through which we see a situation. An 'objective' description, therefore, is one that considers the many different ways that different people see a given situation from their own viewpoints, as well as an explicit acknowledgement of the writer's own lens.
iv) The terms 'qualitative' and 'quantitative' have different meanings in the context of social science. For engineers, 'qualitative' denotes data that is non-numeric. This would include data from what social scientists would regard as a 'quantitative' survey, in which, for example, participants would respond to multiple-choice questions or rate their agreement with certain propositions on a Likert

scale. For social scientists, qualitative data usually refers to open-ended text collected in interviews or observations of human activity. Qualitative data also can include observations about documents, images, drawings, or even natural objects that have been placed in certain positions by people.

Social science theories can provide engineers with a deeper understanding of human behaviour. In particular, a deeper understanding of learning and teaching can make life much easier for an engineer. Our observations of expert engineers underlined the significance of almost entirely self-taught abilities related to teaching. By acquiring some familiarity with social science and education theories, you will be able to get there much faster.

This chapter introduces you to some ideas of teaching and learning that, according to our research, appear to be highly relevant to engineering practice. When you can master them, much of your engineering practice will become significantly easier. The aim is not for you to become an expert teacher, but rather to understand many of the key ideas behind effective teaching. You will learn to identify when teaching performances are needed, who needs to learn, what they need to learn and be able to do, and how you can verify that learning has occurred. If you follow the major references and practice techniques, carefully observing the results, you will find that collaboration performances in your engineering practice become easier and less stressful.

As an engineer with these abilities, people are more likely to do the right thing for you the first time around. They are more likely to trust you and they are less likely to need supervision. You can get more people doing what you need them to do, yet at the same time enrich your understanding by building on the knowledge that other people possess that may exceed your own.

THEORY AND CONCEPTS

Why were you asked to learn so much theory in your engineering course at university?

The answer is simple: theory helps you learn faster, as long as the theory can help explain observations relevant to the task!

In one of the many experiments conducted by learning researchers, psychologists found that students who understood the concept of the refraction of light found it much easier to hit a submerged underwater target using a crossbow while standing by the side of a swimming pool. After one or two trial shots that went wide of the mark, these students realised that you have compensate for light refraction at the surface of the pool. Students who had not learnt about the refraction of light accomplished the same task but took many more attempts before they figured out that you have to aim the crossbow well below the target.[6]

That is a simple example of why it is important for you to begin to understand some of the background theories that can help you learn about teaching techniques.

Some engineering frustrations

Most of the most commonly reported frustrations we found among engineers stem from difficulties in teaching. Engineers often express frustration in these terms:

'I can't get other people to understand simple engineering ideas.'

Here is a senior software engineer who was running a major project at the time.

'We have to make sure that the client's people understand how our systems will help them in their business. It's so easy for their ideas and ours to diverge, for our people to lose track of their needs and you end up with a system that produces little real value for the client. It's a constant struggle.'

Here, you can see two teaching performances. The engineer had to work with the client's people and educate them on how the software could improve their business, while at the same time he also had to learn about the client's business in order to appreciate how the software would help. Then, he had to work with his own software engineers and educate them about the client's requirements, not just once, but continually throughout the project. Without direct contact with the client's people, the software engineers would easily lose track of the requirements, and without continually being monitored and taught, would end up writing software that would not produce a useful outcome for the client. These two performances turn out to be very common in many branches of engineering, particularly in terms of ensuring that the technical intent survives sufficiently intact through multiple reinterpretations by the people who work on implementing it in a way that produces lasting value for end users.

Another example comes from a software engineering R&D team leader who discussed the challenges he had in persuading a group of mining engineers that their software tools needed to be dramatically improved.

'The software engineers can see a bigger picture and advantages that an improved software tool would bring, while our very much overpaid mining engineers and the management group are very satisfied with what they already have and are not eager to change. We prepared a proposal advocating changes and gave examples of other companies that have already adopted better software tools and gained better performance. We became so frustrated trying different strategies and the outcomes. We spent so much energy and time gathering a team of researchers to develop ideas and improve the existing software tools at hand. This proved to me that being technically strong in one dimension can be very good but it's even more important to be an excellent communicator, listener, and a good negotiator. Ability to explain ideas and facts and convincing people to follow is the key of success in any industry.'

DISCOVERY AND TEACHING

As Chapter 7 briefly explains, collaborative learning performances can be classified into two main categories. First, there are situations in which an engineer is learning from other people, but it is reasonably clear that no one has started with a well-defined idea of what needs to be learnt. I have called this performance 'discovery', because learning mostly takes place in the context of working together and involves informal conversations and encounters between people. Oftentimes, the learning emerges almost by accident. Distributed cognition is closely related to this, when new knowledge emerges from social interactions between people.[7]

Teaching, on the other hand, describes performances in which there is a relatively clear idea about the ideas and information that needs to be 'appropriated' by learners through the teaching methods of an engineer.

As we shall see, most effective teaching performances rely on some discovery, which is why I have chosen to include both in this chapter.

So, how do you go about executing an effective teaching performance? How can you convince other engineers that they need to adopt different techniques in order to improve their performance? How can you ensure that other engineers, contractors, technicians, fabricators, and plant operators interpret your ideas faithfully and conscientiously? How can you be sure that your client actually understands what you are proposing to build for them?

Before describing some of the more productive approaches that are useful in discovery and teaching performances, I need to describe some of the important ideas about human learning that have emerged in the last few decades.

Let's start with a common misconception among young engineers. By now, having read Chapter 7, you should no longer have doubts about this.

Misconception 15: A concise and logical explanation is sufficient

Students and novice engineers often think that . . .	Research demonstrates that (particularly for engineers) . . .
A technical idea can be conveyed to another person with a concise and logical explanation.	Bringing other people to a usable understanding of a new idea is difficult, time-consuming, and can be very stressful. Logical explanations are just the start of a lengthy process that involves many steps, takes time to prepare and deliver, and requires careful monitoring to ensure that the new understanding is influencing behaviour in the way that was expected.

Young engineers can be forgiven for this misconception; it reflects the way that they were taught. Engineering students are among the brightest and most eager to learn at any university. They have performed very well at school and are therefore among the easiest young people to teach. Very often, a concise and logical explanation is sufficient for them to grasp an idea and try it out for themselves. Even if this doesn't work, most engineering is taught in this way. The curriculum is dominated by lectures, even in tutorials, during which the scientific principles behind engineering are explained with a few worked examples. The rest is up to the students; those that pass are clearly able to learn a sufficient amount in this way.

There is another important factor: motivation. Students are motivated to learn because they want to graduate and that means gaining as many marks as possible for the least effort. Now, consider the mining engineers mentioned in the quotation on the first page. They are already, apparently, well paid, and improving their performance with new software is unlikely to increase their pay. Also, it will mean that they have to learn something new, so their job will become (at least temporarily) more difficult. Unlike when they were students, there may be little motivation for them to learn.

This is why teaching in the real world is so much more difficult than simply providing concise and logical explanations. Engineers often have to work with people who think they know it all already, and who usually know enough about learning to remember that it is difficult and to be avoided if possible.

Practice concept 51: Planning: the learning objective is a performance

You will need a script.

A teaching performance requires careful thought and planning to avoid frustration. Given that you are working with people, frustration can happen, even with the best thinking and planning beforehand.

Teaching has to start with the idea that the learner or learners are going to have to accomplish a performance with the help of their learning. It may be giving you approval to do something or it may be that they are going to implement some of your ideas in the form of new artefacts or systems. Either way, a teaching performance starts with an assessment of what the learners need to be able to perform and the skills or knowledge that they will need to do that.

We refer to this as the learning objective. A teaching performance starts by defining learning objectives, and also how you, as a teacher, can assess the subsequent performance by the learner or learners to gauge the extent of learning that has occurred.

In other words, what has to be learnt and how can we tell that the learning goal has been achieved?

Practice concept 52: All learning is based on prior knowledge

Students and novice engineers often think that . . .	However in reality . . .
People learn by listening or watching and memorising what they hear or see.	People build new knowledge by adding to ideas they already have. People 'discover' new ideas in terms of the ones they already know about. Anything that does not fit in with prior knowledge may be completely missed.

We learn by listening to other people, watching them, reading books, and so on. Yet, in order to discern something that needs to be learnt, we have to understand what other people are trying to convey to us in words or symbols. As explained in Chapter 7, there is no such thing as uniform understanding of words and symbols. We all have different life experiences and memories; in other words, we all have different prior knowledge. This is especially true when you are working with adults in the workplace, rather than in a formal school or university education setting. Therefore, any explanation will be understood in a way that is unique to every learner.

Our language knowledge is what enables us to perceive something that we need to learn.

Again, as explained in Chapter 7, we may all share English as a common language, to a greater or lesser extent, but we all use English in a different way. Engineers have a special vocabulary: some examples are listed in the glossary (online appendix).

However, it is not quite as simple as that. Behind our language, we all have different ideas about reality. We have a different set of mental models and concepts to explain the world around us. When we perceive something new, we try and fit this perception into our existing mental models.

The introductory chapter of the book, 'How People Learn',[8] illustrates this idea using Leo Lionni's children's story *Fish is Fish*.[9] A baby minnow (a tiny fish) in a pond is anxious to explore the world outside. Unable to leave the pond, the fish befriends a tadpole that eventually grows into a frog. The frog leaves the pond for a while, exploring the world outside. Eventually, the frog comes back and describes what he has seen. The book's illustrations show the way the minnow interprets what the frog has told him: cows appear as fat fish with four fins touching the bottom of the pond, with a large pouch (the udder) between the fins of the back-end. Birds appear as fish with extended fins looking like wings. People are imagined to be fish walking on their tailfins. Although it is not in the book, the frog might have had some difficulty explaining rain: what is the equivalent of rain for a fish? We learn from this simple story that each person receives a story about something new in terms of things they are already familiar with. Therefore, if you can tell the story in terms of familiar words and ideas, it is more likely to be understood.

Susan Ambrose and her colleagues in the book, *How Learning Works*,[10] describe how prior knowledge has to be activated, sufficient, appropriate, and accurate for learning to take place. This poses three challenges: first, we need to know about the language used by the learner, and second, we need to know about the ideas already in the mind of the learner upon which we can build. Finally, we need to ensure that the prior knowledge has been activated and brought to the foreground of the learner's mind.

Let's reflect for a moment on how we learn about the places we live and work and how we move around in a rural or urban environment. Everyone has a different way to represent their understanding of their environment, the mental maps they use to find their way around. For me, my map of the city in which I live closely corresponds to a printed street directory or, these days, Google Maps. As I walk or drive through the city, I retain a reasonably clear idea of the landmarks and districts on either side of my route. If I take a wrong turn, I still have a clear enough idea of where I need to go, so I find my way by driving in the appropriate direction even though I may not know the actual streets I need to follow to get there. I may have to ask for directions or check the street maps once or twice.

My wife, on the other hand, remembers a route as a series of streets that take her from home to her destination. She has no clear idea of the direction in which she is travelling or what lies on either side of her route. She just knows that this route will get her to where she wants to go. While I feel comfortable taking an alternate route, partly to explore another district of the city, she can become completely lost as soon as she leaves her regular route.

Many Aboriginal people in Australia who still live in their traditional 'country' learn how to traverse what seems to us to be endless featureless scrub or even desert by learning songs or stories. Each stage of a song or story is related to a particular place, often involving the names of specific types of plants that grow there. With tens of thousands of different native plant species, many that are found in different forms depending on the soil where they grow, combinations of plant names provide

the equivalent of a street directory for the Australian landscape. The songs or stories string places together to form a route in a way that can be remembered and passed on orally from one generation to the next.

What we learn from this is that we all have different ways of representing the world around us in our memory. As we get to know the area where we live, we gradually build our own mental representation in a way that is related to our culture and traditions. We learn by incorporating new experiences into our mental representation. A teacher cannot simply feed the map directly into your memory; you have to construct a map yourself in your own particular way. A good teacher, therefore, facilitates that learning or mental construction process. An effective teacher helps the learners accomplish this challenging mental construction task for themselves.

Now, think about the way that you remember the way around your local area. Do you remember the appearances of different places, houses, or shops and use them to recognise where you are? Or do you use the street names to work out where you are? Do you look for major landmarks, hills, or tall buildings? Do you think about the angle of the sun during the day to work out the direction you are heading? We are all different, which means that we each have our own combination of techniques.

Now, imagine that you meet a stranger, someone who has only been in the area for a short time. The stranger has asked you for help to find their way to the local branch of a particular bank. How would you begin to explain? (You may not even know where the local branch of that bank is located if it is not the bank that you use, but let's assume that you do know.) You might begin by asking the stranger how he reached you, and what else he knows about the surroundings. 'Have you seen the tall B&OD building, the one with blue-coloured glass? No? Well, have you seen the huge Shell garage, the one with the McDonald's next door?'

If the stranger speaks broken English with many grammatical mistakes, you will quickly realise that they may not understand English very well. You may have to adapt the way you speak in a way that makes it easier for this stranger to understand, namely by using simple words and speaking slowly. In other words, you are thinking about the stranger's grasp of language, and how to provide directions in a way that the stranger is more likely to be able to understand.

Having confirmed what the stranger already knows about the district, you might then remind them about a particular landmark needed to find the bank. 'Okay, now walking from the Shell garage past McDonald's, you go past three or four shops, and then you will find a jewellery shop on the left-hand side, just on the corner. Have you seen that before?' Having confirmed that the stranger has seen the jewellery shop, you can then direct them along a side street that leads to the bank.

How would you know if the stranger now has an accurate idea of the route to the bank? The stranger might be trying to understand what you are talking about and is not really quite sure about the jewellery shop, but is confident that it can easily be found since it is quite close to the McDonald's outlet. How sure can you be that the jewellery store is still there on the corner? How many shops do you have to go past? Are you sure that it's just three or four shops? Or is it actually several hundred metres along the same street? Maybe the stranger cannot actually remember the large Shell garage next to the McDonald's outlet but does not want to seem completely ignorant about major landmarks in the district. With any of these slips, the stranger will construct new

knowledge about the district that is incorrect and misleading. As a result, the stranger may become lost and have to ask for directions again.

Does it really matter if the stranger has an accurate idea? With enough determination the stranger can gradually find the way to the bank by a combination of people pointing in the right direction, as well as simple trial and error.

Returning to the four prior knowledge conditions that have to be satisfied in order for learning to take place.

i) Appropriate: the stranger knows about landmarks close to the bank branch.
ii) Sufficient: the stranger knows about the Shell garage, the McDonald's outlet, and the jewellery store.
iii) Accurate: the stranger can remember which side of the street the jewellery store is on.
iv) Activated: through conversation, the stranger has reminded himself or herself of these landmarks.

In engineering, teaching performances matter. Facilitating learning by others that is right the first time means that you spend less time and energy having to check later and correct mistakes or misunderstandings by other people.

Finding out about prior knowledge is not as easy as it seems. Merely asking someone if they already know about reverse cycle air conditioners, as an example, is not going to help you. Some people will respond, 'Yes, of course I do'. These people do not want to give the impression that they are ignorant about something as basic as an air conditioner. Then there are others who will say, 'No, I don't'. These people might be concerned that you will then go on and ask them something about how air conditioners work: they don't want to be shown up as being ignorant by being unable to answer your follow-up question. Therefore, being cautious, they deny any knowledge.

Here is an example from an engineer in the oil and gas industry. She found that understanding her client's prior knowledge was crucial before trying to educate them about the engineering work that needed to be done for their projects.

> *'Most of our real clients know a lot about engineering issues and they have their own engineers. However, they often don't know much about construction issues. Often they have a focus on operations and maintenance without understanding how construction affects that. The less experienced clients often are not aware of regulatory issues. For example, if they had not recently built some hydrocarbon process plant they may not be aware that you need to alert the Department of Environmental Protection at least 12 months ahead or their construction work will go nowhere. In subtle ways you need to ask questions to figure out how much they are really aware of. For example, you can casually bring AS1210 (Pressure Vessel Design Code) into the conversation and judge from their responses whether they know about it in detail or not.'*

As this example shows, casual conversations are more effective at exposing prior learning. Try starting a conversation about a topic that is likely to be of interest to the learner: it might be about a recent sporting match or a political incident reported in the media. Gently move the conversation to the topic of recent hot weather and how many people

are using air conditioners, perhaps by posing a rhetorical question, such as, 'I wonder what kinds of air conditioners are used around here?' A casual conversation like this can help to activate prior knowledge; in other words, the prior knowledge is brought closer to the surface of consciousness. Particularly with adult learning, when you have no real idea of what the other person already knows, accurately understanding prior knowledge is indispensable. You need to be able to assess whether it is sufficient or appropriate. It might be quite inappropriate or it may simply not be enough to comprehend the ideas that you are going to try and teach.

Now think about a situation that you might face as an engineer when you need someone to perform some skilled technical work. For example, imagine that you have received a special purpose hydraulic control valve from an equipment supplier and you need to have it checked out by technicians to make sure that it functions as expected. You will need to provide technicians with instructions on what to do: this is a teaching performance.

At first you seem to have a clear idea of what needs to be done, but when you think about it, there are many detailed aspects that you are not sure about, especially if you are not a mechanical engineer or you are not familiar with hydraulic control systems. For instance, a technician will need to connect the control valve to a hydraulic circuit with the appropriate instrumentation. Is there a suitable testing facility already in the workshop? How will the valve be flushed out to remove any small particles of dirt that could interfere with its operation? Does another hydraulic circuit have to be adapted for the test? Is there instrumentation with the required measurement accuracy? What tests will need to be performed? You may know the answers to some of these and the technician will probably have answers for some of the other questions.

Most likely, you will start by arranging a time to meet with a senior technician and discuss the testing process together. If you have not worked with the technician before, you will spend time figuring out how much the technician actually knows about these kinds of control valves and different testing methods. You may not be very sure about the kinds of tests that need to be performed if you have not worked with similar control valves before.[11] This is typical of an engineering discovery performance, similar to the discovery performance between you and the stranger asking directions to the bank: you are trying to establish some understanding of prior knowledge. At the same time, you may be listening carefully to the technician to pick up some of the language that they are using. You may have no idea of what is meant by a Swagelok fitting for example, so you admit to that ignorance and subsequently learn about a common type of pipe connection used in hydraulic systems.

Once again, think about how you would confirm that the technician's prior knowledge satisfies these requirements.

i) Appropriate: the technician is familiar with hydraulic testing methods and constructing hydraulic circuits.
ii) Sufficient: the technician knows enough to understand about the testing work you need done.
iii) Accurate: the technician has up-to-date knowledge of the actual components needed.
iv) Activated: through your conversation or recent work on similar hydraulic circuits, the technician can quickly access the required knowledge.

That's why teaching almost always starts with a discovery performance where we engage the learner in a casual conversation about ideas that might be relevant for future learning. However, it is more than casual for the teacher; you will be paying particular attention to the words that the learner uses and the ideas expressed in those words. These observations will form the foundation on which you are going to plan your teaching. You would remember from Chapter 5 about the importance of taking notes and of recording the actual words being spoken. Any message that you want the learner to comprehend has to be expressed using the words that are already familiar in the learner's vocabulary.

In more formal settings, particularly working with several learners at the same time, there may be large variations in prior knowledge between them. As an engineer, you might encounter this when starting up a new project or working with a supporting team of other engineers and support staff. You need to educate them on what's been done so far and what needs to be done for the next phase of the project.

If the aim of the learning is to develop conceptual knowledge, it can be helpful for learners to construct graphical representations of their existing knowledge: sketches, concept maps, plans, or other diagrams that represent their prior knowledge. The advantage of a graphical representation is that you can often quickly appraise the extent of their prior understanding by glancing at a diagram, rather than reading an extended piece of text.

A more useful technique in vocational learning situations is to ask the learners to engage in an activity that will require them to use their knowledge. It is useful to be able to set it up as an entertaining game so that everyone can have a few laughs. However, the game has a serious intent: you get to watch the learners perform. From that, you can judge their prior knowledge. You can even set up a mock judging panel composed of some of the participants and get them to score the others' performances. Obviously, you should rotate participants through the panel so that everyone gets a chance to evaluate others.

It is important to make sure that people are in a comfortable environment for the discovery aspect of your performance. In many situations, sharing a meal together can be a great way to start, because it provides plenty of opportunities for casual conversation. The more listening you do and the more talking your learners do, the more you will be able to discover about what they already know and understand. Equally important, by listening carefully, you may be able to gauge what they *don't* understand. In this book, many student misconceptions about engineering practice have been pointed out. All these have come from listening to students and young engineers talk about engineering practice, writing down what they said, and then thinking about the ideas that might have led them to make those statements.

Any of these techniques could be appropriate for working with the technician in the example above. Think about how best to appreciate the extent of the technician's knowledge, and your own, and the gaps that might have to be made up by learning or experimentation, another discovery performance.

Once you understand what your learners already know, then you can work out what they need to learn. With adult learners, a good way to do this is to engage them in further discussion – see if they can suggest what they need to learn. Later, we will discuss learner motivation. Adult learners who have some say in deciding what they need to learn are much more likely to be motivated. This is particularly appropriate

when collaborating with skilled technical workers. Asking them to figure out what needs to be done demonstrates your respect for their knowledge and your desire to learn from them.

COLLABORATIVE DISCOVERY PERFORMANCES

As explained earlier, a discovery performance is one in which the learner engages in conversation with other people and learning happens as a result of that conversation. The conversation may be face-to-face, over the telephone, or even in writing. Neither the learner nor the others involved have a clear idea of what the learner needs to know; those details emerge gradually through the discussions. A discovery performance can last anywhere from a few minutes of a single casual conversation to several months that involve a whole series of meetings and encounters.

Take the example of the engineer-inventor working with the patent attorney. The engineer has a clear idea of the invention but needs to work with the patent attorney who knows how to write in the way that is required for a patent specification. Initially, the patent attorney has only a brief written description of the invention. After sitting with the engineer who can explain it to him, the attorney develops a better idea and then works from the documents to produce the first draft of the patent. Later the attorney would ask the engineer to check the patent specification to make sure that the specification explains all the important elements of the intervention and the reasons why they have been incorporated. Since the intended reader of a patent specification is a similarly skilled engineer or technician, the patent attorney may ask questions on how other engineers might interpret the words of the specification. Gradually, after working with an attorney on several inventions, the engineer develops a clearer idea about how patents are written and may even be able to produce a draft patent specification that requires only a minor amount of further editing. However, the attorney is always more likely to be aware of legal technicalities and the necessity to use certain techniques in drafting, particularly for the critical claims usually listed at the end of the patent.

Here is an excerpt from an interview with an engineer talking about document review, another example of a discovery performance:

> *'Once an engineer produces a document, then it is checked by another engineer who goes through and redlines[12] the document. He checks every line, and also has a checklist that he has to tick off. Then it would come to an approver – that's me. I add my comments on to it and then it's updated by the originating engineer. Then the documents are circulated for review and approval. Typically I'll get them on the review cycle. I have my opportunity to make changes. But the way it works – because it's kind of multidisciplinary – the subsea people will get it, also drilling, operations and other groups. They see it as well. They put in their comments and we'll take their comments into account if we think they are appropriate. At the end of the day, it's up to me if we take them or not, as an approvalist.*
>
> *'[When a document gets redlined, what sort of issues are picked up?][13] You pick up all sorts of stuff from just grammar, bad writing, punctuation, all that kind of normal stuff. Plus obvious questions about functionality. We find mistakes in*

assumptions – not often with pressures, though. But gas composition, temperatures potentially. You know, that kind of stuff. So you get to check it for technical accuracy as well. The other issue is where there has not been good communication between engineering disciplines. You find a missing issue somewhere that you are aware of but the author of the document has not been made aware of it and has written the document without taking it into account.'

The person speaking has to check the work of the less experienced engineers writing documents, such as specifications. The document author is learning as a result of seeing the comments written by other engineers and discussing them with his supervisor. The reviewer learns from the author as he reads the document: the author will have investigated technical issues in much more detail than the reviewer has time for. Engineers in other disciplines learn about developments elsewhere in the project team as they read the document, looking for improvements that have to be made with respect to their own discipline area. Finally, as explained in the last sentence, the reviewers also learn about communication gaps in their own organisation, as well as possibly learning about weaknesses on the part of the author.

The final example demonstrates yet another kind of discovery learning performance for engineers.

'We asked the equipment vendors to do some engineering work and that required a budget, I guess a couple of million dollars each. We had to sell that to the vendors and our management. It was actually to do some engineering, prior to them preparing their proposals. In a way, it actually funded their proposals. What happened was that we wanted to get a better idea of what they thought about what the equipment should look like and what challenges they foresaw. We paid them to kind of pick their brains, if nothing else. I think it was money well spent. I think our management did as well, at the end of the day.'

This engineer is talking about a recent experience in which he and his colleagues had to choose between different vendor firms supplying some highly complex equipment for a special project. This engineer had extensive experience but knew that the engineers in the firms had much more specialised knowledge than he had been able to acquire himself. In some instances, particularly if there is fierce competition between the firms, the suppliers may have been willing to allow the engineers to spend considerable time developing and explaining their proposals. Engineering firms normally see the education of prospective users of their products as a very high priority and part of their business development (or marketing) efforts. In this case however, at the instigation of the engineer being interviewed, the customer firm has recognised the importance of going further than normal. They have paid a substantial amount of money to commission the vendors' firms to perform a more detailed study of their requirements, partly so that their own engineers could learn from the vendors' engineers before starting to evaluate the vendor proposals.

Notice how all of these examples demonstrate how knowledge is distributed among different people in the overall engineering enterprise, as explained in Chapter 5. They also show how discovery learning takes time, which has to be carefully balanced against the likely benefits.

Table 8.1 Some discovery performances in engineering.

Asking another person about the advantages and disadvantages of competing products or services, learning about future developments in competing products or services.
Posing a technical query in order to clarify what seems to be missing information or a mistake.
Seeking to understand more about other people or stakeholders and learning about their opinions, viewpoints, languages, values, and interests.
Ask about progress that has been made on a piece of technical work.
Negotiating shared meaning, a common understanding, taking diverse perspectives into account, aligning stakeholders – all of which have been shown to predict successful collaborative problem solving.
Discussing the need for a particular artefact, solution, or process with people such as end users or sales staff for whom it might provide value to assess their level of interest and their needs.
Working with a patent attorney to prepare a patent specification: a claim for the limited monopoly rights of an inventor for a particular innovation. The engineer understands the invention, but not how to write a patent, whereas the patent attorney understands how to write a patent, but needs to learn about the invention from the engineer.
Arranging for technical work to be reviewed by peers, possibly external to the enterprise.

Effective learning from discovery performances

Table 8.1 is framed around the likelihood that you, the reader, need to acquire much of your learning in discovery performances. How can you use the previous discussion on learning to make this easier for you? Can you identify your prior knowledge in a way that could help you make better use of a discovery learning opportunity? After the performance has finished, you could try and identify where your prior knowledge was insufficient or included misunderstandings.

To make the most effective use of a discovery learning performance opportunity, it is a good idea to prepare with some self-study. Start by reading documents that will be the focus of the discussion or closely examining other relevant artefacts. Practice your drawing skills in order to notice details that might otherwise escape your attention. Above all, identify technical terms that might be unfamiliar or used in different ways. You may need to clarify the meanings during the discussions. Don't overdo preparation either; remember, this is a learning experience. Be prepared to ask many questions, preferably open questions that encourage other people to talk about what they know. For example, if you are exploring how much someone already knows about air conditioning technology, it is not a good idea to ask a question like, 'Is that the liquid trap?'

This is an example of a closed-form question: the answer is simply yes or no. Instead, an equivalent open-form question might be:

> *'This cylinder here, I'm not sure, but it looks like it could be a liquid trap to me, could you tell me about different arrangements that you may have seen and how they compare to this one?'*

This kind of question encourages the other person to talk more, giving you the opportunity to learn from their experience and observations. If, on the other hand, the other person has little technical understanding, the way that you have posed your question, acknowledging your own uncertainty, allows them to be open about their own uncertainties by saying something like this.

'Yes, it might be a liquid trap, but I haven't noticed things like that before.'

Practice your listening skills.

Make sure that you take notes. It's not easy to ask people to remember what they said in these discussions and people don't necessarily repeat themselves when you want them to. Be prepared to share your notes and ask the other participants to check them for technical accuracy later.

Make sure that you remember to learn about the ways in which people use different words. Don't let unfamiliar words, or phrases used in an unexpected way, slip by without understanding how the speaker intends the meaning to be received. It may be different from your understanding of what the word or phrase means. It might seem awkward to be repeatedly asking people to explain what they mean by a particular word or phrase in a meeting. Try to remember that everyone else has been where you are and probably remember their own feelings of ignorance, their reluctance to ask questions, and their regret at not having asked questions when they had the opportunity. Also, more experienced engineers may be even more reluctant than novices to expose their own uncertainties, so they may be very grateful that you asked for clarification.

Ensure that you write notes as people provide explanations, recording the actual words used as accurately as possible. Although this takes time, you will gain the respect of others. The time and trouble that you take to prepare notes demonstrates your respect for their time and trouble in providing you with explanations.

TEACHING PERFORMANCES

Table 8.2 lists several teaching performances in engineering. In reality, however, it is sometimes hard to distinguish individual teaching performances and separate them from discovery learning performances. Real life is inevitably more complex, as different people learn from each other. Sometimes, the teacher learns just as much as the learner, if not more.

Since most literature on teaching and learning focuses on formal classroom learning in formal educational institutions, it is helpful to examine a real case study drawn from engineering practice. The engineer speaking to us in this case study is both a learner and a teacher. She is responsible for a critical component in a process facility using high-pressure, high-temperature fluids that pose a significant safety hazard if any leaks or equipment failures occur. The actual words of the engineer have been edited to make the case study easier to read and to protect the identity of the people involved.

'This special high-pressure high-temperature reactor vessel, I was the initiator of the concept. It was very frustrating initially because the company's ears were closed to that idea, based on prejudice, I think, from other projects, I don't know.

Table 8.2 Some teaching performances in engineering – refer to text for explanations about the columns listing attributes of the performances.

Performance description	*Learner*	*Intent*	*Timing*	*Mode*	*Meeting?*	*Follow-up*
Convey, specify the details of the intent behind the design of an artefact or process so that others can manufacture it with the predicted attributes or operate the process successfully.	performer	implicit	later, synchronous	writing/f2f	y	inspect, monitor
Provide a technical explanation: a verbal performance with the aim of building a level of technical understanding in the mind of another person or audience.	performer	implicit	synchronous	f2f	y	monitor
Advocate the particular advantages of an artefact, solution, or process with the intention of convincing another firm or engineer to purchase it or recommend it for procurement.	potential client	implicit	synchronous	f2f	y	follow-up
Write a report to be placed on record: a succinct statement summarising a technical issue to record what was known and by whom at a particular point in time.	engineer, inquirer	intentional	historical	writing	n	none, self-evaluation
Write a request for quotation: a detailed description of an artefact, information package, service, or material to be purchased, written to enable suppliers to prepare quotations.	potential partner	intentional	later	writing	n	responses
Write a quotation: a firm estimate of the cost to provide specified information, an artefact, a defined quantity of material, or to follow a defined process.	potential client	intentional	later	writing	n	follow-up
Write an expression of interest: a response to an enquiry from an enterprise seeking to procure an artefact, information package, service, or material, indicating that your firm is interested in being approached.	potential client	intentional	later	writing	n	follow-up

Perform a demonstration: a performance in which an artefact or other equipment or materials are demonstrated with the aim of reassuring the audience about the capacity to achieve a given level of performance in service.	decider, peers	intentional	later, synchronous	f2f/video	y	responses, follow-up
Inform: enabling or empowering other people to make use of information.	info user	intentional	later, synchronous	writing/f2f	y	monitor
Help another person to learn how to perform a task or how to do something to help achieve desired objectives.	performer	intentional	synchronous	f2f	y	monitor, follow-up
Provide guidance and reassurance to help a less experienced person develop their technical capability.	performer	intentional	synchronous	f2f	y	monitor, follow-up
Reassure a client that the project will be completed on time and within budget with the required technical performance.	client	unintentional	later, synchronous	writing/f2f	y	responses
Warn: alert someone to a technical issue and the need for pre-emptive action to avoid undesirable consequences.	decider, peers	unintentional	later, synchronous	writing/f2f	y	monitor, follow-up
Stake out or assert your territory: helping people understand your competence and expertise and ability to take responsibility for a given aspect of a technical undertaking.	peers	unintentional	later, synchronous	writing/f2f	y	monitor, responses
Advocate a particular technical approach.	decider, peers	unintentional	synchronous	writing/f2f	y	monitor, follow-up
Influence perceptions: conversing with another person in a way that encourages the person to change their perception about other people or of a technical issue.	decider, peers	unintentional	synchronous	f2f	y	monitor, follow-up

I have been focusing on justifying the decision to use a reduced wall thickness than is normally required,[14] *justifying that the risk is acceptable to the operator, I guess. There are still some doubters within the company that it is going to work, ultimately.*

'I am supervising some reliability modelling being performed in London by specialists. I had to develop the detailed methodologies for that, discussing it with the experts in London and also the standards authority. It is a step-by-step process – we develop the concept and we have that subjected to third-party review by the standards authority. I had to visit Norway and London, just bringing the parties together to reach consensus. Now they are doing the modelling[15] *and the next step will be to review the results and present those to the company again, and then have those subjected to an independent external review yet again.*

'I have to make sure that we don't lose sight of what we are trying to achieve. Technical integrity[16] *is essential to keep the standards authority people in Oslo on side. Then we have to bring along many other stakeholders*[17] *within the company particularly the operations people – they are very nervous about the whole thing. It has always been quite a challenge.*

'I don't have all the technical competency I need, myself. I can't understand all of the nuts and bolts around structural failure mechanisms and stress modelling – it is a very specialist area of statistics and engineering that you just hope that the guys that are doing it know what they're doing.

'What we are doing is we're really extending these methodologies – we are not changing the fundamentals. So in that respect, I guess, that's given me comfort that I am not completely out on a limb in terms of proposing this solution.

'[You have daily telephone conversations on this?] Yes – what we have now are a slight misalignment is in the modelling results between our own work by the specialists and other work that the standards authority did in the past. So at the moment we are just trying to define the process, step-by-step, by which we can hopefully rationalise the differences.

'I have to manage the politics of that because we have two sets of professional people who probably both think they are right. I need to attempt to find some middle ground that we can both agree on. Perhaps somewhere along the line we've missed a piece of detail in our reviews of the methodologies and that might explain the misalignment. When we do the numbers, they have come up different. Why the difference? We are starting to understand that – it wasn't expected.

'Ultimately it is the probability of failure which you can then hopefully believe and then you can apply your margins to that number, inherently applying conservatisms in methodology. We are just demonstrating, we believe robustly, that we have a huge safety margin so we don't need those extra millimetres of thickness. That leads to a significant change in the project economics.

'[And you're working on this more or less by yourself?] That's right though I have another engineer, one of our own designers who acts as a sounding board for me. He is a very competent engineer so I use him as a reviewer as well.'

This engineer had realised early on that reducing the thickness of the metal reactor vessel walls by a few millimetres would significantly reduce the cost. In this particular

case, with an exotic metal required for the reactor vessel, the cost saving was very large. There were further cost savings than just the material cost: with reduced thickness, the welding process would be more reliable. With a thicker wall, many of the joints would have to be bolted, requiring even more material and high-quality machining operations.

In proposing this design, however, this engineer had stretched the interpretation of the Norwegian standard, which caused concern among her colleagues and senior management. At the centre of this lies a teaching performance; the engineer has to convince her senior management and colleagues, particularly those colleagues responsible for the plant operation, that the design is safe and the probability of failure is acceptably small. The particular standard in question is based on statistical modelling because of the large number of welded joints needed for this complex containment vessel, as well as the fact that many vessels are required.

In order to provide a more extensive justification for his proposal, the engineer had already convinced her senior management to commission an independent expert in London to perform complex mathematical modelling. The engineer had to educate these experts on the complex design of the reactor vessel and the need to be able to perform calculations in order to demonstrate that, even with a reduction in wall thickness, the strength of the reactor vessel would be much greater than required for this application. However, after all the work of setting up computational models, these experts have produced results that differed from earlier studies conducted by the standards authority in Norway.

Notice how this engineer used the term 'politics' in the sense of dealing with disagreements between highly expert engineers in different organisations. Engineers often use this term to refer to difficult situations when opinions differ for reasons that they don't necessarily understand. It is not that they have performed the same calculation and obtained different results. Both have used slightly different mathematical and computational approaches, with different assumptions, and each group of engineers has different prior experience. Their interpretation of the technical issues is therefore different, as you would now understand given the earlier discussion in this chapter, as well as Chapter 7.

The way to resolve this is to engage in a discovery learning performance: an extended series of telephone calls, meetings, and e-mail correspondence, in which all of the actors, including our engineer, gradually learn about their different viewpoints and learn to see the same situation from a different perspective.

Rachel Itabashi-Campbell, in her study of problem solving by engineers in the American automotive industry,[18] revealed similar teaching performances that she referred to as 'aligning stakeholder perceptions of the problem'. She showed how engineers have to spend a lot of effort on this performance. Success in aligning stakeholder understanding, however, was the main factor that distinguished successful problem-solving efforts from unsuccessful ones.

Practice exercise 16: Identifying teaching and learning performances

Take a moment to read the case study quotation above once again. See how many discovery, learning, and teaching performances you can identify in the text. (A self-evaluation guide is provided in the online appendix.)

Variations in teaching performances

Table 8.2 lists a selection of teaching performances commonly encountered in engineering practice. Apart from the description, the other columns list the attributes of each performance.

The 'learner' column describes who is going to be the learner: it is taken for granted that the engineer is the teacher.

The 'intent' column indicates whether the learning will be implicit, intentional, or unintentional. Intentional learning occurs when the learner sets out to engage with a person or document in order to learn something. Implicit learning occurs when the learner engages with the teacher or document produced by the teacher, even though the learner did not necessarily have the intention to learn or the expectation of learning. The act of engaging with the document creates the implicit expectation that learning will occur, even though the learner may not see it as a learning exercise. In several performances, the learning is unintentional. It happens without a desire to learn on the part of the learner. In other performances, the learning is intentional. The learner engages with the teacher or documents in order to improve their knowledge or understanding.

The 'timing' column indicates whether the learning will be synchronous with the teaching as it is in face-to-face encounters or will occur later, possibly without any interaction with the teacher. This book is an example of the latter case; although it might be possible for you as a reader to interact with me by e-mail, I hope that this book is sufficiently self-explanatory in the context of your engineering practice that we don't need to correspond in order to me to clarify ideas in the book. In the case of a technical report, it may be prepared in case it is needed at some stage in the future recording details of engineering work or testing performed for some purpose. The report may never be read by anybody other than the author and those who checked it. The report may even be read as a historical document many years or decades later.

The 'mode' column indicates whether the interaction between the teacher and the learner will be face-to-face (f2f) or through written documents, possibly with e-mail correspondence.

The 'meeting' column indicates whether there will be at least some interaction during the performance in the context of a formal meeting.

The 'follow-up' column indicates how the teacher can confirm that learning has taken place.

Several of performances could take place in modes other than those shown in the table: it is not the intention that the table should include every possible engineering teaching performance.

The first entry in the table might be the most familiar to a novice engineer, who has been educated with the idea that the solution for a problem or a design needs to be communicated to a client who will then use the results of the analysis. As we have seen earlier in this book, engineers rarely, if ever, directly participate in the production of artefacts or delivery of services resulting from their work. Therefore, they can only influence what happens by conveying the technical intent emerging from their work so that other people can perform the work with that knowledge to guide them.

Remember this quotation from an engineer with long experience in maintenance organisation?

> *'No amount of reliability modelling calculations serves any useful purpose until a fitter uses his tools in a different way.'*

Therefore, the challenge for the reliability engineer is to find an effective way to influence the fitter's actions. To do that, the engineer needs to convey technical intent to the fitter and others involved in maintenance.

Most young engineers would imagine all that is needed is an e-mail with appropriate instructions and perhaps a drawing or two. Just because the documents exist, of course, does not mean that people actually read them, let alone act on the contents. That's what teaching performances are all about.

Learning is not easy; it takes considerable physical energy on the part of the teacher and the learner, as well as a significant amount of time. Furthermore, it is not a one-way process as is so often portrayed by the simplistic notion that many young engineers hold, that teaching is simply a transferral of information from the teacher to the learner. Instead, as we have seen earlier, learners add new knowledge by building on their existing knowledge, just like the way that we extend the mental map of our surroundings as we visit places for the first time. If we extend our mental map on the basis of missing or incorrect information or associations, we get lost and need to ask for help. In most learning situations, it is rare for the learner to construct new knowledge without needing some help and guidance to resolve differences of interpretation. The teacher has to observe the learner perform in a way that reveals both correct and incorrect understandings. By reinforcing the correct understandings, the learner gradually develops workable knowledge that will enable an appropriately good performance.

Given the effort needed for any kind of learning and teaching, it is not surprising that motivation plays a big part in securing the required engagement and interest of the learner. Looking at Table 8.2, we can see that many teaching performances involve unintentional learning. These pose a significant challenge, because the learner does not expect to be learning and is not necessarily willing to devote the required effort. As such, understanding how to motivate and engage learners is another prerequisite for effective teaching performances.

Practice concept 53: Value, interests, expectancy, and environment: motivation for learning

In addition to figuring out learning objectives, it is also important to work out why your learners need to learn. Adult learners, in particular, are very strategic about their learning. They only learn when they know they need to. Adults can also be faster learners than young people because they don't bother to learn what they don't need. Young people, on the other hand, often don't have a clear idea of what they will need in practice, so they attempt to learn everything that they can. Unfortunately, they don't always learn it very well.

With formal learning in an educational institution, assessment provides the most powerful motivation for learning. Every student needs sufficiently high marks to pass. Learners soon adapt their learning behaviour to achieve those marks in order to reach their specific objective – to pass, to do well, or to excel. Informal learning has no assessment mechanism, however. Therefore, other motivational factors become much more significant for influencing learning performance.

Learning consumes considerable cognitive energy. Physiological studies suggest that the brain consumes around one third of our body's energy production. It certainly consumes more when it has to learn, and studies suggest that intense emotions, physical

exercise, and recovering from infection or injury all compete for our body's supply of energy. That's why motivation is so important: motivation helps direct our energy towards learning.

Learning science has pointed out two critical factors that contribute to the motivation to learn: *subjective value* and expectancy.

Subjective value and interests

Researchers have identified three sources of *subjective value* that drive learning.

- *Attainment value*: the internal and personal satisfaction you gain from mastery and accomplishment of a goal or task, even if no one else notices.
- *Intrinsic value*: the satisfaction gained from simply doing the task itself. Personally, I always used to find building models much more satisfying than completing them; once they were built, I put them away in a cupboard and often never looked at them again. For me, model building had great intrinsic value. Corporate values come into play here. Learning that aligns with corporate values is more likely to provide intrinsic satisfaction.
- *Instrumental value*: the degree to which achieving a goal will help you progress towards achieving later more important goals or achieving extrinsic rewards, such as praise from others, public recognition, money, material goods, an interesting career, or a high-status job. This is where interests become important; demonstrating that the learning has a positive impact on the interests of the learner helps to establish instrumental value, and hence motivation.

One of the most useful ideas here is to identify the principal interests of the learners. Interests are measures of outcomes, either qualitative or quantitative, that provide greater benefits (or reduced penalties) for a person or a group. Many people have an interest in earning more money. Some people are more interested in earning a predictable amount of money, rather than earning more, but having it come with more income security. Some people may have an interest in building relationships, others in experiencing something new and interesting. Many engineers have an interest in learning about new technologies and see this as something that might make one job opportunity more attractive than another.

If the teacher can demonstrate to the learner that their interests will be advanced as a result of learning, the learner is much more likely to be motivated.

One of the best ways to identify the immediate interests of the learner is to start with a simple question: 'What is it about this situation that could be keeping you awake at night?' The ensuing conversation is likely to provide you with a rich understanding on the most immediate interests, or other motivational factors, that will drive learning. It may well tell you that the learner is preoccupied with issues that are completely unrelated to the planned learning, in which case it might be better to wait for another opportunity.

As a teacher, you need to clearly link the effort that learners need to devote to these three subjective values. Demonstrating your own *intrinsic value* is a good way to start; 'Hey everyone, I think this is really cool!' Of course, not everyone is going to find everything intrinsically satisfying, particularly if it turns out to be difficult. Helping people see the *instrumental value* always helps; 'This will be really helpful when you

have to diagnose faults with electronic sensing equipment.' Notice how this book is structured around the idea that you can become an expert engineer. By explaining why the ability to teach is valuable for expert engineers, I have provided you with *instrumental value* to help motivate you to learn this material.

Practice exercise 17: Identify your own interests

See if you can identify your own interests with respect to your current employment situation. If you are not working in an engineering job at the moment, what are the interests that would help determine your choice to work with one company or another. Is the location important? Would the prospect of travel be important to you? How much do you expect to get paid? Do you expect to be paid if you have to work more than the normal number of hours in a day? Is the opportunity to work with advanced technologies important to you? Are you seeking stable employment with job security? How much do you value training opportunities? Do you prefer technically challenging work, or are you more inclined to prepare yourself for a management career? Are you interested in developing a reputation for technical excellence, making it difficult for you to compromise when a quick and clumsy solution is adequate?

Expectancy

Expectancy is the second element of motivation. Students need to think that they can achieve the learning goals. In this regard, it is particularly powerful for a student to say: 'Hey people, if I can even get this, then we all can.' Note the use of the 'we' that identifies the speaker as one of the group, rather than a separate teacher who might say 'you'.

One of the best illustrations comes from my own experience. When I started engineering, the dean of the faculty told us in the first lecture to look at the person sitting to our right and our left sides. 'Only one out of the three of you will make it through the course,' he said. And that's what happened. The course completion rate in the engineering faculty at that time was about 33%. This was a self-fulfilling prophecy; any student who was doubtful about their ability to complete the course had their doubts resolved at that moment: they would fail. And they did. Psychology research has now demonstrated that everyone has the potential to achieve any learning goal they set themselves. The only major difference between people is the rate at which they can learn. Some people simply learn faster than others.

It is important to keep reinforcing a simple message: 'We can all do this, together.' Keep repeating that!

The 'together' word is important. Learning takes place within a social context and the people around us help create the motivation we need to learn by giving us encouragement.

Environment

The last factor that contributes to motivation is a supportive environment. This has nothing to do with the strength and durability of the building. However, it has everything to do with the encouragement and emotional support that we provide to others.

Table 8.3 Learning motivation responses.

	Unsupportive environment		*Supportive environment*	
	Little subjective value	*High subjective value*	*Little subjective value*	*High subjective value*
Expectancy (efficacy) is HIGH	Evading	Defiant	Evading	MOTIVATED
Expectancy is LOW	Rejecting	Hopeless	Rejecting	Fragile

Here's an example of an unsupportive teacher, or even a fellow learner: 'Why can't you get it? Any idiot can understand that!' Here is a much more supportive remark: 'I can see that it's been a struggle to get your mind around this, and I really appreciate you devoting time and energy to understand all this.'

Think about ways in which you can establish the right combination of motivation – high expectancy, high subjective value, and a supportive environment.

It is very important to pay attention to the actions and attitudes of learners. Listen to what they are saying to each other. If you misinterpret what is needed or how they are progressing, you will end up with learners who reject your guidance, evade your teaching, and deliberately do not listen to you. Take a look at Table 8.3.

A major part of environmental support can come from organisational and community values. Here I am using the term 'values' in a different way: it is not the same as the plural of 'value' mentioned earlier.

In this context, values are either stated or unstated behavioural attributes derived from actual behaviour that people respect as being virtuous, which shape the way individuals perceive their environment, relate to other people, and make choices within a community, an enterprise, or an organisation. The relative importance of different values for an individual may contrast sharply with those of another, however.

Here are some examples of values:

Forgiving – recognising mistakes and offences as almost always unintentional
Kind, generous – giving to others without expectation of favours in return
Caring, honest, responsible, trustworthy, fair, firm – but not unyielding
Disciplined – not easily distracted, perhaps difficult to manipulate
Flexible – responsive to needs of others, possibly easily distracted
Status conscious – aspiring to be one who is admired by others and whose opinions one respects
Stable, predictable – feeling of security, often in the form of material possessions or money, avoiding uncertainty

Recent research in economics has helped to demonstrate how powerful values can be in motivating collaborative behaviour within an organisation – far more powerful than financial inducements. Most people devote more effort to their work with an organisation with which they have shared common values. Leading companies have recognised this and devote considerable resources to create statements of their corporate values that most of their workforce can identify with.[19]

IBM, for example, recently decided on a set of corporate values for the first time since the company was founded. Most major companies display their values on a web page that can easily be found using search terms 'corporate values' or 'company values'. Here are some examples:

Delighting our customers
Acting as a single team
Acting with integrity and respect for others
Being safe and environmentally responsible
Delivering value and profit

Companies tend to choose values that other people positively identify with, for example, helping other people, courage, determination, and honesty. Values are powerful determinants of action when faced with choices. For example, when faced with the choice of completing a task as quickly as possible in order to leave work by a given time, or staying behind and completing the task to a high degree of quality, care, and attention, what choice would you make? Some people value conscientiousness and work more highly than family time, and others place more value in spending time with their families. However, by reminding everybody about the company's values, which might include doing everything possible to keep customers happy, people are a bit more inclined to devote extra care and time to their work, knowing that the result will make somebody happier.

Of course, we all have different values and the extent to which we have shared values with the company or organisation will be different from one person to the next. However, by describing the need for learning in terms of the company values, it is more likely that people will understand the need in positive terms and thus devote a little more effort than might have otherwise been the case.

It is important to realise that the espoused values, the values that appear on the company website, may not be well aligned with the enacted values, which are the values that influence the real decisions that people make in the workplace. For example, most engineering companies strongly emphasise health and safety in their corporate values. Unfortunately, the strength with which health and safety appears to be emphasised in corporate values does not always reflect the reality in the workplace. For example, in a televised interview after the catastrophic explosion of the BP Horizon rig in the Gulf of Mexico in 2010, the chief operating officer of BP, Bob Dudley, stated that the company strived for the right balance between safety, business, and performance.[20] In other words, he was saying that the business performance requirement for economic production of oil and gas required compromises on safety. He was reflecting, no doubt, the enacted company value on safety, overlooking the reality (obvious in hindsight) that without safety, there can be no production from a rig that has blown up and sunk to the seabed. In spite of its immense size, he acknowledged that BP came extremely close to complete financial failure as a result of the explosion and the subsequent leakage of oil that caused extensive pollution in the Gulf of Mexico. The company had lost the confidence of investors and suppliers that it could meet its short-term financial obligations; the severity of the global community's mistrust of the company following the disaster was intense.

There are some values that emerged from our research as ones that many engineers subscribe to and identify with. When working with engineers, you may be able to use these to gain an extra degree of engagement. Notice that some may conflict with organisational interests (if not values).

Technical satisfaction – getting the technical solution right for the job and having enough time to confirm it as such rather than having to move on because of budget constraints.
Emergence – seeing plans and ideas take shape and go into effect, artefacts emerge from countless discussions, drawings, designs, and documents.
Elegance – a solution that is simple and aesthetically pleasing from a technical standpoint.
Impact – a perception that one's work has made a positive difference to the outcome that can be pointed out to one's family: 'I designed that' or 'You see that small yellow bit, that resulted from my contributions, if they hadn't had it. . . .'
Innovation – novel or inventive in a significant way, or in advance of what peers would consider 'state-of-the-art'.
Challenge – work which tests one's abilities, technical and/or organisational, which nevertheless is just within one's level of confidence and competence, albeit requiring an advance in learning or discovery relative to one's abilities or knowledge at the start to be satisfactorily completed.
Humanitarian – work that has evident social benefit, particularly for disadvantaged people in poor communities.

One way that you can build on these values to gain extra engagement from engineers is to offer ways for engineers to have more experience with work that they can identify as having these attributes. Again, not all engineers subscribe to these values; there are large individual differences.

Practice exercise 18: Find your company values

Find out the values upon which your company is founded. If you are not working, choose any company that you associate with engineering activities and search for the value statement on their website. Compare the espoused values with the enacted values that you observe in the workplace, or, if you are not employed, research the company's environmental and safety performance from news reports and compare this with the values that they claim.

If you can, carefully consider each of these values and work out the extent to which your own values align with them.

Practice concept 54: Delivering the message – more than being an interpreter

Now, after reviewing the preparatory discovery performances to learn about the languages, prior knowledge, conceptual understandings, and motivations of the learners, we can begin to construct the message. The ideas that we would like the learners to appropriate[21] may have emerged in the context of an entirely different social

community, using different language, concepts, and ideas from those familiar to the learner.

Therefore, your challenge in a teaching performance is, in at least one sense, to be an interpreter. You may be trying to explain to a mechanical designer why electrical cables in a robot manipulator need to be constrained to keep them from bending with less than a minimum allowable radius of curvature or you may be trying to cross a much wider gap in understanding. You may be trying to explain to an accountant why it is wiser to purchase the most expensive seals for a hydraulic cylinder, rather than the least expensive ones with a shorter operating life.[22] You may need to provide directions for a cement truck driver who has minimal proficiency in English. Wet concrete cannot be stored and it can be a huge embarrassment if the truck driver delivers it to the wrong location on a construction site. You may be working on the physical layout of high-frequency electronic circuit components and need to explain to a mechanical injection-moulding designer where certain components need to be placed to minimise electromagnetic interference or emission susceptibility.

Remember that most of the engineers you will work with will not have read a book like this. Many, but not all, focus their interests and work in a relatively narrow specialised domain of interests and knowledge. As an aspiring expert engineer, your challenge is to build on their knowledge, skills, and perhaps long experience. However, you can only do this if you translate and reconstruct your needs and requirements into a frame of reference that is familiar to them. You don't need to match their technical understanding of their particular engineering sub-discipline. However, you will need to understand enough of their prior knowledge to be able to describe what needs to be done in terminology they understand and through the use of their concepts.

By explaining how learning helps them with their immediate interests, you are more likely to keep the learners sufficiently engaged to do the hard work that is required to master a new concept.

Why is this more than a normal interpretation performance, translating the message into language that the learner will understand?

First, it is not just a matter of words. The concepts and ideas may be quite foreign to the learner. Therefore, you need to return to the language and ideas of the learner that you discovered (and noted down accurately) earlier. Instead of simply translating the message into the learner's language, you will find yourself reconstructing the message to build on ideas that are familiar to the learner. Instead of asking the cement truck driver to position his discharge chute to pour cement onto the lift shaft footings (part of a building under construction, familiar to the engineers who have an understanding of every detail of the drawings), you might get into the cab with him and show him where to back up the truck for the first concrete delivery. To the truck driver, a construction site ready for a concrete pour is just a mass of holes in the ground with steel reinforcing bars sticking up all over the place.

Thus, the essence of the actual teaching performance is to construct a meaningful message for the learner by using the words and ideas that are familiar to the learner in a way that addresses immediate interests. You can only do this if you have taken the time to listen to the learner and identify the best approach for teaching.

There are many helpful books that can help you deliver a message effectively once you have crafted the content with sufficient understanding about the prior knowledge

and ideas in the minds of your learners. However, many of these books overlook some of the barriers that you need to take into account when delivering your message.

Your learners may not be able to listen accurately in the ways that you have learnt earlier in this book. If they take notes, they might retain between 5% and 50% of what you say, and then only if you have their complete attention. Go back and review the chapter on listening skills and remember the cues you can watch for that will tell you if your learners are listening. If not, stop, reassess, and avoid wasting their time and yours. Find a way to regain their attention and focus: 20–30 seconds of silence is not a bad way to try and do this.

Delivering the message in writing is more challenging because you will not be present to answer questions or help with clarification when you see your learners having difficulties.

Engineering students are less accustomed to reading than students in other academic disciplines. From data collected during my teaching career, I have learnt that only 10% of my engineering students read the most important written material that I provide in my courses: the paragraphs on assessment requirements that specify how their marks will be calculated. The rest still know what was in those paragraphs; they come to know which students will have read them and ask them to tell them about the assessment requirements. Only about 2% read material that does not seem immediately relevant.

Do not assume that engineers presented with your written material will have changed much since they were students. Assume that most of them will only read if there is no other way to find what they need. Assume that they will miss a significant proportion even when they do read the material, so remember to repeat the most important messages more than once.

Put yourself in the place of an engineer who is a reluctant reader and needs to find the really important material in a hurry, material that is written in a way that makes sense to an engineer. Many engineers feel more comfortable with diagrams, even if they only serve as convenient place markers showing where relevant writing is located.

Variation method

In its essence, the variation method for constructing a message is so simple that it seems obvious. To explain an idea, think of its intrinsic variations, and then draw the attention of the learner to the part that remains constant through all the variations. It sounds easy, but it is not nearly as easy as it looks.

Here's a simple example. To teach a child the colour purple, we show the child several objects that are all coloured purple, maybe with slightly different shades of purple. We might say 'Henry, this is a purple car, this is a purple tube, this is a purple bottle top, this is a purple plate, etc.'. However, it is all too easy not to notice other similarities. Just by coincidence, we might have chosen four or five plastic objects to teach 'purple' to Henry. However, they might also all have been more or less round (e.g. a plastic toy car that is round, with round wheels). Then, without us realising, Henry thinks that anything that is made of similar plastic material is purple. We didn't notice this because we already had 'purple' in our minds and simply grouped a bunch of purple objects that happened to be nearby.

In their paper, 'On Some Necessary Conditions of Learning', which is challenging yet highly rewarding reading, Marton and Pang demonstrate not only the extraordinary effectiveness of teaching using this basic technique, but also how easy it is, even for highly prepared and experienced teachers, to get it wrong without realising.[23] It takes great care, experience, and insight to achieve an effective teaching strategy.

Zone of proximal development

Let us return once again and remember the brief discussion on teaching the stranger directions to a particular bank branch. Imagine for a moment that the way to get there is rather complicated and involves 25 stages, with the route marked by 30 or more landmarks. Unless the stranger had developed the capacity for total recall of verbal instructions, it is unlikely they would remember more than the first two or three steps of the route. That might be sufficient; the stranger would then stop and ask someone else for directions.

This thinking exercise illustrates the idea of a zone of proximal development.[24] This loosely refers to the extent of learning that the student is capable of at one time, with the help of other learners and the teacher. No matter how hard we try, an individual can only learn a certain amount in one sitting. Everyone is different, of course, and it is not easy to figure out how much anyone can learn at a particular time. However, the principle behind this is important. Gradually, when teaching, we develop a sense of what is possible to learn at one time.

In practice, what it means is that you have to think about the limitations of the learners. If you are going to try and explain something that involves too many conceptual steps and new ideas, it is not going to happen in one session. Therefore, you need to think about time constraints. You may need to build a 'learning bridge', a way of reaching a useful learning outcome without having to learn all the intermediate details that might seem important to you. This means thinking carefully about what you want to learn to be able to do at the end of the process. It may be necessary to simplify the background knowledge required, so that it is just enough for them to perform adequately. Later, they can learn more of the background details that might provide a more in-depth appreciation. Think about your own experience through school and university. You may have had to study similar material many times, maybe twice in primary school and again twice at secondary school and then again at university. Each time, you learnt the subject in more detail and emerged with more insight than before. However, for many purposes, the primary school learning might have been quite sufficient. It got you to where you needed to be quickly.

I faced a similar challenge in writing this book. Our research interviews and field observations yielded around 5,000 pages of text. Putting all the relevant research literature together (books, journal articles, conference papers, and reports) contributes perhaps 15,000 pages of text. Even though this book will run to about 600 pages when it is finished, my aim has been to provide you with as much learning as is reasonably possible from those 20,000 pages of text without losing too much valuable information. I have tried to highlight the important concepts and also misconceptions that can impede learning. In doing so, I have had to construct many conceptual bridges and borrow many more from literature to provide you with something from which you

can learn. Also, I hope that you will remember that most of your learning will take place when you learn to interpret the world around you in terms of the ideas in this book.

Collaborative learning

The attraction of cooperative learning in an engineering setting lies in the similarity with distributed cognition processes that occur naturally in the workplace, as discussed in Chapter 5 page 140. Engineers adopt the same kinds of interactions as have been observed in cooperative learning classrooms. Cooperative learning has been demonstrated to be more effective than most other techniques in hundreds of carefully controlled trials.[25]

Cooperative learning[26] describes a large family of methods in which learners help each other learn. There are both formal and informal learning methods. Hundreds of carefully controlled trials have demonstrated large learning improvements compared with conventional teaching methods. (Note that the term 'co-op education' is used in the USA to describe internship or workplace learning. This is simply a clash of terminology; there is no direct connection with cooperative learning.)

Smith et al. (2005) provide a concise description: 'Formal cooperative learning groups are more structured than informal cooperative learning groups, are given more complex tasks, and typically stay together longer.' Well-structured formal cooperative learning groups are differentiated from poorly structured ones on the basis of the characteristics presented in the table. From these characteristics we can distil

Table 8.4 Effects of structure in cooperative learning.

Less structured (traditional)	*More structured (cooperative)*
Low interdependence. Members take responsibility only for themselves. Focus is typically on a single product (report or presentation).	High positive interdependence. Members are responsible for theirs and each other's learning. Focus is on joint performance.
Individual accountability only, usually through exams and quizzes.	Both group and individual accountability. Members hold self and others accountable for high-quality work.
Little or no attention to group formation (students often select members). Groups are typically large (4–8 members).	Deliberately formed groups (random, distribute knowledge/experience or interests). Groups are small (2–4 members).
Assignments are discussed with little interest in each other's learning.	Members promote each other's success. They do real work together, helping and supporting each other's efforts to learn.
Teamwork skills are ignored. Leader is appointed to direct member participation.	Teamwork skills are emphasised. Members are taught and expected to use collaborative skills. Leadership roles are shared (by rotation, for example) among all members.
No group processing of the quality of its work. Individual accomplishments are rewarded.	Group processes (discusses) quality of work and (evaluates) how effectively members are working together. Continuous improvement is emphasised.

five essential elements in successful implementation of formal cooperative learning groups: positive interdependence, face-to-face promotive interaction, individual accountability/personal responsibility, teamwork skills, and group processing.

MASTERY

Learners don't develop mastery simply by listening to an explanation.

At the early stage of learning, every task can be difficult, even listening to the introduction. The student is consciously trying to understand new ideas in a new context, which requires significant energy. Even elementary tasks require the brain to think in ways that it has not tried before. The brain literally needs to develop new neural connections; this takes time and energy.

With practice, the new connections strengthen and become persistent in the brain. We reinforce the right connections through feedback. The *pleasure from achievement* literally provides chemical signals that reinforce the most recently used neural connections.

Practice makes perfect but only if the learner receives informed and reliable feedback. Without reliable feedback, practice can result in inappropriate skills and connections becoming reinforced and later they can be difficult to undo.[27]

These ideas can be summarised with the learning pyramid, figure 8.1. The diagram is attempting to show how the essence of good teaching is to allow the learners to discover new knowledge for themselves. This is particularly true in the company of other people who provide both a *supportive environment* and also the *extrinsic value* that comes when others recognise the learner's achievement.

Most people only learn effectively when they use new knowledge in the company of supportive fellow learners. Some people can learn very effectively by themselves. They have less of a need for recognition from their peers and teacher. They gain more subjective value from the *intrinsic pleasure* of doing it for themselves and the *attainment satisfaction* that comes from completing learning tasks successfully. When they do attend classes, these individuals can be very helpful, because they often can provide additional support for students who thrive in a classroom environment.

Checklist for effective teaching performances

As explained earlier, you will need a script. By working through this checklist, you can develop your own script to your discovery learning and informal teaching performances.

Identify teaching and discovery performances that will be needed

Listing all the separate teaching and discovery performances will help you appreciate the tasks that lie ahead of you.

Identify the learners and outline the learning objectives. For each learning objective, work out how you will be able to confirm that the required learning has taken place.

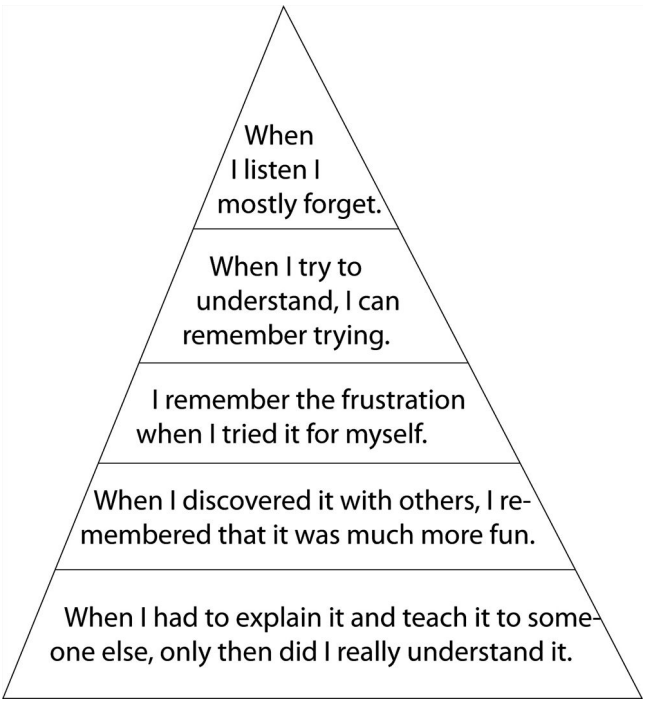

Figure 8.1 Learning pyramid – each of the comments reflect increasing degrees of learning from the top to the bottom. The bottom statement emphasises the power of cooperative learning in which each participant has as much responsibility for teaching the others as learning for themselves.

Discover the learner's language and prior knowledge

Confirm that prior knowledge:

i) is appropriate,
ii) is sufficient,
iii) is accurate, and
iv) has been recently activated.

Ensure that you listen carefully to learners to develop an understanding of their language and terminology. Engage learners in casual conversation and try out some of the learning messages, carefully listening to the responses.

Ensure that you can construct the learning messages using terminology and language that is both familiar and comfortable for the learners. Ask peers who are fluent in the language of the learners to evaluate your messages ahead of time in case adjustments are required.

Characterising teaching performances

Assess the degree of learning intention by the learners: will they participate in the performance knowing that they are there to learn something? Or will the learning be incidental, without any intention to learn?

Assess the timing: will the learning take place in a face-to-face environment in your presence? Will some or all of the learning take place later, without you being present? Will the learners be able to interact with you and ask questions?

What will be the medium for instruction? Face-to-face, the use of artefacts, electronic mail, websites, and documents can all be used singly or in combination.

Will the learning take place in the context of a formal meeting, or at least be supported with formal meetings?

Will the learning be monitored and evaluated by the inspection of documents, or by monitoring specific performances by the learners?

Motivation: values and interests

Identify specific interests of the individual learners and develop an appropriate script that builds on these interests, particularly to highlight the instrumental value of learning.

Ensure that the script identifies the attainment value, intrinsic value, and instrumental value of learning.

Expectancy and environment

Identify ways to build a supportive learning environment: one that creates the expectancy among the participants of sufficient learning to achieve their objectives.

Ensure that the script contains messages to promote positive expectancy, using 'we will . . .' rather than 'they will . . .' so that the teacher is placed in the community of learners and is not separated or placed on an authoritative pedestal.

Ensure that your script contains hints and suggestions on creating a supportive learning environment.

Identify corporate values so that your learners may share and develop an appropriate script that links motivation with these values.

Presenting the message: more than interpretation

What are the ideas and words that are familiar to the learners with which you can construct the message they need to appropriate?

Do they need to see objects to understand? Did they work with sketches or images to explain their ideas to you, hand signs? Did they take you to the actual places to show you something or could they describe it in words effectively?

Construct the message using familiar words, places, ideas, and images. Do you need to introduce them to a new idea? What might already be familiar to the learners that you could use to explain the idea?

Can you use the variation method, illustrating what is constant through all possible variations in terms of context and background?

Zone of proximal development

Carefully consider the learning objectives and, with an understanding of prior knowledge among the learners, decide whether you need to construct 'conceptual bridges' so that useful learning that leads to satisfactory performances by the learners is achieved in the required time.

Attention span

Construct a learning schedule that recognises limitations on the attention span of the learners. For most people, it is wise to keep the attention requirement under 15 minutes for a single session. Ensure that your script enables you to summarise the learning objectives in 30 seconds.

Rehearsal and recording

Like any other performance, a rehearsal is likely to greatly improve the quality, particularly if the rehearsal takes place in front of a friendly audience. Even before the rehearsal, practice your teaching performance many times in your own room, imagining that there is an audience sitting in front of you. The more often you do this, the more confident your presentation will become.

The more often you can practice your explanations, even to the extent of boring your friends and acquaintances, the more likely you will be able to deliver a polished performance when it is really needed.

Record your presentation, especially for yourself, so that you can criticise it and improve it later, but also so that you know exactly what you said. You can also make the recording available to others if they need it later.

Monitoring learning

Watch for learners nodding their heads or saying, 'Yes, I understand that', which indicates that the learners think they have learnt something. However, what they have learnt might be quite different to what you expect.

Only by carefully listening to their spoken responses and subsequent actions can you determine whether any actual, useful learning has taken place after the session is over. This requires the teacher to follow-up with discovery performances similar to the ones required to assess prior knowledge. This will enable a teacher to find out whether the learners have changed the way they think about what they have learnt, or whether their perceptions have appropriately changed. If the aim was a skilled performance by the learners, it will be necessary to observe the performance to see whether the learning has had a legitimate effect.

As we shall see in the next chapter, follow-up and monitoring requires significant amounts of time from engineers.

Mastery

How can your learners practice to further develop learning? Remembering that effective learning through practice requires a means of evaluating performance, work out how learners may be able to evaluate their own performances. If this is not possible, further teaching or mentoring may be essential.

SUMMARY

Most young engineers assume that other people learn the way they did at school: the teacher presents a logical explanation, perhaps using a set of PowerPoint slides, and that is enough to solidify the concept.

By now, you probably realise that teaching – being the interpreter – is anything but simple.

However, more than anything else, effective teaching lies at the core of expert engineering practice. We shall see in the following chapter how critical teaching is for the most common collaboration performance for engineers – technical coordination.

Thanks to recent research, mostly since the start of the 1990s, we now have some powerful techniques that can help you learn how to teach effectively. There is not space to introduce them all here, but the references will lead to you other effective techniques.

Once you have read this chapter, find some friends who are schoolteachers and discuss the ideas explained in this chapter with them. You may be amazed how much you can learn from them that will positively impact your own approaches to sharing knowledge and teaching others.

How will you be able to tell whether your discovery learning and teaching performances are improving? If you have improved your discovery performances, you will have more comprehensive and accurate notes to refer to, with keywords and ideas highlighted so you can use them for teaching. When you present your message to your learners, they will quickly show you that they have understood, and may even thank you for explaining something in language they can easily understand. They will echo your message accurately. While this does not show they can remember it accurately, at least you know you have mastered the first part of a teaching performance.

When you have improved your teaching performances, you will notice how much less time you spend correcting mistakes made by learners and, yet again, showing them how to do it right. You will spend less time fixing up the consequences of mistakes and incorrect actions that happened because people did not act in the way you expected them to.

Don't expect your skills to show definite improvement every time. I teach students year after year and every cohort is different; they react in unexpected ways and the explanations that seemed to work every time before somehow lead to confusion. It might be something small that you have done differently, or it might simply be that the learners are different from the previous group.

The important thing is to keep trying. With the material in this chapter, you should be able to outperform most other engineers, until they catch up with the help of the lessons in this book.

NOTES

1. Ashforth, Sluss, & Saks (2007); Eraut (2004, 2007); Fuller & Unwin (2004); Hager (2004); Katz (1993); Korte (2007); Marsick & Watkins (1990); McGregor, Marks, & Johnston (2000); Wenger (2005)
2. Wenger (2005).
3. Engeström (2004); Rooney et al. (2014); Trevelyan (2007).
4. Bailey (2010) Personal communication, Also see further details reported in Bailey and Barley (2010).
5. Martha Nussbaum(2009) has refocused attention on the capacity to understand a situation from the point of view of someone else, and this has become a significant issue in recent discussions on design. See for example Oosterlaken and Hoven (2012). Lucena and

Leydens have also drawn attention to this capability in their notion of contextual listening (Lucena & Leydens, 2009).

6. Bransford et al. (2000).
7. Itabashi-Campbell et al. (2011); Lam (1996); Nonaka (1994).
8. The introductory chapter of the book *How People Learn* describes the view of learning in which new knowledge is constructed on foundations provided by previously learnt knowledge (Bransford et al., 2000).
9. Lionni (1970).
10. The book *How Learning Works* provides references with links to contemporary research in education psychology and neurological science (Ambrose et al., 2010).
11. Chapter 5, Figure 5.10, shows a map of specialised technical knowledge.
12. Each company adopts its own convention for marking up changes and corrections, such as using red lines to indicate errors in assumptions or logic, green for improvements to English expression, etc.
13. Square brackets denote a question by the interviewer.
14. Thickness of the metal walls of the pressure vessel.
15. Modelling to predict the likely behaviour of the pressure vessel when subjected to varying pressures and temperatures causing additional stress due to thermal expansion and contraction, coupled with the possibility of stress-induced corrosion affecting the strength of the welds joining the metal parts.
16. Integrity here means the process by which engineers assure themselves (and external reviewers) that an installation will meet its design requirements through construction and its anticipated service life in operation.
17. Stakeholders are people or groups of people who can influence a project or whose interests could be influenced by the project. In this context, they would include the engineer's management team, senior managers in the company, the likely vendors supplying the reactor vessel, the London experts and their management, the operations engineers and their management, and other members of the project team.
18. Itabashi-Campbell & Gluesing (2014); Itabashi-Campbell et al. (2011).
19. Akerlof & Kranton (2005).
20. ABC Four Corners, March 11, 2011. For an extended discussion, refer to the PhD thesis written by the chief investigator for Piper Alpha, Exxon Valdez, and the BP Horizon blowout disasters (Bea, 2000).
21. Appropriate is a verb used in education psychology that refers to the multiple mental processes that lead to the development of new knowledge, understanding, skills, and capabilities in the learner.
22. Seals, relative to the overall cost of machinery, are relatively inexpensive items. However, if they fail prematurely, they can be very expensive to replace because of the time required to disassemble and reassemble complex hydraulic machinery. More expensive, longer life seals are likely to be much more cost effective than cheaper ones.
23. Marton & Pang (2006).
24. The concept originated in the 1920s from the work of the Russian child development psychologist, Lev Vygotsky (Bransford et al., 2000; Brown et al., 1993).
25. K. A. Smith, Sheppard, Johnson, & Johnson (2005).
26. Felder & Brent (2007); D. W. Johnson & Johnson (2009); D. W. Johnson & Johnson (1999).
27. Draper (2009a, 2009b).